Metric Manifolds

Nishtha Tikalal

1 Motivation

So far a smooth manifold carries no way to measure anything: no lengths of vectors, no angles, no lengths of curves. A connection (Lectures 7–8) tells us what "straight" and "parallel" mean, but not how long a curve is or which curve is the *shortest*. This lecture adds the missing structure, a *metric*, and then shows that it is not independent of the connection: a metric determines a canonical connection.

2 Metrics

Definition 1 (Metric)**.** Let $(M, \mathcal{O}, \mathcal{A})$ be a smooth manifold. A *metric* on M is a smooth $(0,2)$ tensor field g such that, at every $p \in M$,

(i) $g_p(X,Y) = g_p(Y,X)$ for all $X, Y \in T_pM$ (symmetric),

(ii) the map $T_pM \to T_p^*M$, $X \mapsto g_p(X, \cdot)$, is an isomorphism (non-degenerate; equivalently, $g_p(X,Y) = 0$ for all Y implies $X = 0$).

A manifold with a metric is a *metric manifold* $(M, \mathcal{O}, \mathcal{A}, g)$. In a chart the components $g_{ij} = g(\partial_i, \partial_j)$ are smooth functions forming a symmetric, invertible matrix at each point; the inverse matrix is written g^{ij}, so $g^{im}g_{mj} = \delta^i{}_j$. Since g is a tensor, the g_{ij} transform as a $(0,2)$ tensor between overlapping charts, which is what makes the definition chart independent.

2.1 Musical isomorphisms

The isomorphism in (ii) is denoted $\flat$:

$$\flat : T_pM \to T_p^*M, \qquad X \mapsto X^\flat := g(X, \cdot), \qquad X_i^\flat = g_{ij}X^j.$$

Its inverse is $\sharp : T_p^*M \to T_pM$, with $(\omega^\sharp)^i = g^{ij}\omega_j$. This is what is meant by *lowering* and *raising indices*. The inverse metric is itself a $(2,0)$ tensor, $g^{-1}(\omega, \sigma) = g(\omega^\sharp, \sigma^\sharp)$.

Warning. Writing X_i for $g_{ij}X^j$ is convenient, but X and $X^\flat$ are *different objects* (a vector and a covector). The same letter with the index moved is only meaningful once the metric is specified, and a different metric gives a different covector. Without a metric, indices cannot be lowered.

2.2 Signature

At a point p, g_{ij} is a real symmetric matrix, so it has real eigenvalues. Under a change of basis the matrix changes by $g \mapsto A^T g A$, which can change the eigenvalues but, by Sylvester's law of inertia, not the *number* of positive and negative eigenvalues.

Definition 2 (Signature)**.** The *signature* of g at p is the pair (s,t) (often written as a list of signs) of the numbers of positive and negative eigenvalues of $g_{ij}(p)$; $s + t = \dim M$ by non-degeneracy.

Because the components are continuous and eigenvalues cannot cross zero (non-degeneracy), the signature is constant on a connected manifold. The two cases of interest:

- **Riemannian**: signature $(+,+,\dots,+)$: $g_p(X,X) > 0$ for $X \neq 0$.
- **Lorentzian**: one sign differs, $(+,-,-,-)$ in four dimensions (the opposite overall sign $(-,+,+,+)$ is an equivalent convention). One fixes one convention and sticks to it.

In the Lorentzian case a vector X is called *timelike*, *null* or *spacelike* according to the sign of $g(X,X)$, which gives the causal structure used later in relativity.

3 Length of curves

Let $\gamma : [0,1] \to M$ be a smooth curve with velocity vector v_γ. The *speed* of γ at $\gamma(\lambda)$ is

$$\sqrt{g_{\gamma(\lambda)}\big(v_{\gamma,\gamma(\lambda)},\, v_{\gamma,\gamma(\lambda)}\big)} = \sqrt{g_{ij}(\gamma(\lambda))\,\dot\gamma^i(\lambda)\dot\gamma^j(\lambda)},$$

where $\dot\gamma^i = \mathrm{d}(x^i \circ \gamma)/\mathrm{d}\lambda$ are the components of the velocity in a chart x. In the Riemannian case this is well defined; in the Lorentzian case one takes $\sqrt{|g(v,v)|}$ and restricts to non-null curves.

Definition 3 (Length)**.** The *length* of γ is

$$L[\gamma] := \int_0^1 \sqrt{g_{ij}(\gamma(\lambda))\,\dot\gamma^i(\lambda)\,\dot\gamma^j(\lambda)}\;\mathrm{d}\lambda.$$

Speed is the more fundamental notion; length is merely the integral of speed.

Remark (Units)*.* The components $\dot\gamma^i$ of the velocity have units of (coordinate)/(parameter). The metric components g_{ij} are the conversion factors that turn this into a physical speed (length per parameter): $g_{ij}\dot\gamma^i\dot\gamma^j$ has units of length^2/parameter^2. In a chart whose coordinates are angles, for instance, the g_{ij} contain the radius. It is the metric, not the coordinates, that gives meaning to "distance".

Theorem 1 (Reparametrization invariance)**.** *Let $\sigma : [0,1] \to [0,1]$ be an orientation-preserving diffeomorphism. Then $L[\gamma \circ \sigma] = L[\gamma]$.*

Proof. By the chain rule, $(\gamma\circ\sigma)^{\cdot\, i}(\lambda) = \dot\gamma^i(\sigma(\lambda))\,\sigma'(\lambda)$, with $\sigma' > 0$. Hence

$$L[\gamma\circ\sigma] = \int_0^1 \sqrt{g_{ij}\dot\gamma^i(\sigma(\lambda))\dot\gamma^j(\sigma(\lambda))}\;\sigma'(\lambda)\,\mathrm{d}\lambda = \int_0^1 \sqrt{g_{ij}\dot\gamma^i(\mu)\dot\gamma^j(\mu)}\;\mathrm{d}\mu = L[\gamma],$$

by substituting $\mu = \sigma(\lambda)$. □

So the length depends only on the image of the curve (and its orientation), not on how it is traversed. This is a feature the physics of Lecture 9 does *not* share: there, the parametrization of a world line is meaningful.

Example 1 (The round sphere)**.** On S^2 use the chart (θ,φ), which covers most of the sphere (the angles must be restricted to $\theta \in (0,\pi)$, $\varphi \in (0,2\pi)$, so a meridian is left out and the poles are not covered). Define $g = \mathrm{d}\theta\otimes\mathrm{d}\theta + \sin^2\theta\,\mathrm{d}\varphi\otimes\mathrm{d}\varphi$, i.e. $g_{\theta\theta} = 1$, $g_{\varphi\varphi} = \sin^2\theta$, $g_{\theta\varphi} = 0$. The components are defined chart by chart, and one has to check that on overlaps of charts they are consistent, which is the tensor transformation law.

- The curve $\varphi(\lambda) = 2\pi\lambda$, $\theta = \pi/2$ (the equator) has speed 2π and length $L = 2\pi$.
- A latitude circle $\theta = \theta_0$ traversed once has length $L = 2\pi\sin\theta_0$, shorter than the equator.

4 Geodesics from a variational principle

We now ask for the curves that extremize $L[\gamma]$ among curves with fixed endpoints. By reparametrization invariance we may choose the parameter so that the speed is constant (an *affine* parameter), and then the same extremal curves are found from the action

$$S[\gamma] = \frac{1}{2}\int_0^1 g_{ij}(\gamma)\,\dot\gamma^i\dot\gamma^j\,\mathrm{d}\lambda,$$

which has no square root (its extremals also have constant speed). With the Lagrangian $\mathcal{L} = \frac{1}{2}g_{ij}(x)\dot x^i\dot x^j$, the Euler–Lagrange equations $\frac{\mathrm{d}}{\mathrm{d}\lambda}\frac{\partial\mathcal{L}}{\partial\dot x^m} - \frac{\partial\mathcal{L}}{\partial x^m} = 0$ give

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\big(g_{mj}\dot x^j\big) - \tfrac{1}{2}\,\partial_m g_{ij}\,\dot x^i\dot x^j = 0 \;\Longrightarrow\; g_{mj}\ddot x^j + \partial_k g_{mj}\,\dot x^k\dot x^j - \tfrac{1}{2}\,\partial_m g_{ij}\,\dot x^i\dot x^j = 0.$$

Symmetrizing the middle term, $\partial_k g_{mj}\dot x^k\dot x^j = \frac{1}{2}(\partial_k g_{mj} + \partial_j g_{mk})\dot x^j\dot x^k$, and multiplying by g^{im}:

$$\boxed{\ddot x^i + \Gamma^i{}_{jk}\,\dot x^j\dot x^k = 0, \qquad \Gamma^i{}_{jk} = \tfrac{1}{2}\,g^{im}\big(\partial_j g_{mk} + \partial_k g_{mj} - \partial_m g_{jk}\big)}$$

These $\Gamma^i{}_{jk}$ are the *Christoffel symbols* of the metric. The equation has exactly the shape of the auto-parallel equation of Lecture 8, so the extremal curves look like auto-parallels of a connection with these coefficients, a hint that the metric determines a connection. We now construct it abstractly.

Example 2 (Geodesics on the sphere)**.** For the metric above, the non-zero Christoffel symbols are $\Gamma^\theta{}_{\varphi\varphi} = -\sin\theta\cos\theta$ and $\Gamma^\varphi{}_{\theta\varphi} = \Gamma^\varphi{}_{\varphi\theta} = \cot\theta$, so the geodesic equations read

$$\ddot\theta - \sin\theta\cos\theta\,\dot\varphi^2 = 0, \qquad \ddot\varphi + 2\cot\theta\,\dot\theta\dot\varphi = 0.$$

The equator $\theta = \pi/2$, $\dot\theta = 0$, $\varphi = \omega\lambda$ solves them: great circles are geodesics. A latitude circle $\theta = \theta_0 \neq \pi/2$ does not ($\ddot\theta = \sin\theta_0\cos\theta_0\,\omega^2 \neq 0$), although it is a perfectly good curve.

5 The Levi-Civita connection

Given a manifold with a metric g, there is a *preferred* connection. A connection ∇ is called *metric compatible* (with g) if $\nabla g = 0$. Its geometric meaning: if X, Y are parallel transported along a curve γ, then

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\,g(X,Y) = (\nabla_{v_\gamma}g)(X,Y) + g(\nabla_{v_\gamma}X, Y) + g(X, \nabla_{v_\gamma}Y) = 0,$$

so *parallel transport preserves lengths of vectors and angles between them.* This is the condition that ties the notion of "parallel" to the measuring structure.

Theorem 2 (Levi-Civita)**.** *On a metric manifold $(M, \mathcal{O}, \mathcal{A}, g)$ there exists a unique connection ∇ that is torsion-free and metric compatible ($\nabla g = 0$). Its coefficients in a chart are the Christoffel symbols above.*

Derivation of the coefficients. Let $\Gamma_{jik} := g_{jm}\Gamma^m{}_{ik}$. In a chart, $\nabla g = 0$ reads

$$\partial_k g_{ij} = \Gamma_{jik} + \Gamma_{ijk},$$

and torsion-freeness means $\Gamma_{jik} = \Gamma_{jki}$. Form $\partial_k g_{ij} + \partial_i g_{jk} - \partial_j g_{ik}$ by cyclically permuting the indices in the first relation and using the symmetry in the last two indices of Γ: all terms cancel except $2\Gamma_{jik}$, so

$$\Gamma^m{}_{ik} = \tfrac{1}{2}\,g^{mj}\big(\partial_i g_{jk} + \partial_k g_{ij} - \partial_j g_{ik}\big).$$

This shows uniqueness; existence follows because these are consistent connection coefficients (they transform as required in Lecture 7). □

The auto-parallels of the Levi-Civita connection are exactly the extremal curves of the length functional: *geodesics* of the metric. In a Lorentzian metric a geodesic keeps its causal character (timelike, null or spacelike) all along, by metric compatibility.

Remark. There is a notion of connection without any metric (Lecture 7), and generally a connection and a metric on the same manifold are independent data. Requiring $\nabla g = 0$ and zero torsion is a *choice* that singles out the Levi-Civita connection. This is the connection used in general relativity.

6 Curvature of a metric manifold

Since a metric manifold carries a canonical connection, it has a canonical Riemann curvature tensor $R^i{}_{jmn}$ (Lecture 8). From it one builds:

- the *Ricci tensor*, a contraction of the Riemann tensor, $\mathrm{Ric}_{jn} = R^m{}_{jmn}$, which for the Levi-Civita connection is symmetric;
- the *Ricci scalar*, $R = g^{jn}\mathrm{Ric}_{jn}$, which needs the metric to take a trace of the two lower indices.

The Ricci tensor needs a connection only, while the Ricci scalar needs the metric as well. These are the objects that enter Einstein's equations in gravity (the Einstein–Hilbert action is built from R).

Summary

- A metric is a symmetric, non-degenerate $(0,2)$ tensor field g; it gives isomorphisms $\flat, \sharp$ between vectors and covectors (raising and lowering indices, always with respect to g).
- The signature (numbers of $+$ and $-$ eigenvalues) is independent of the chart and constant on a connected manifold; Riemannian $= (+,\dots,+)$, Lorentzian $= (+,-,-,-)$ (one convention fixed).
- Speed is $\sqrt{g(v_\gamma, v_\gamma)}$ and length is its integral; it is invariant under reparametrization.
- Extremal curves satisfy $\ddot{x}^i + \Gamma^i{}_{jk}\dot{x}^j\dot{x}^k = 0$ with $\Gamma^i{}_{jk} = \frac{1}{2}g^{im}(\partial_j g_{mk} + \partial_k g_{mj} - \partial_m g_{jk})$.
- The unique torsion-free, metric-compatible connection is the Levi-Civita connection; parallel transport preserves inner products. Its curvature gives the Ricci tensor and scalar.